

Proximal Methods in Numerical Optimization

Lecture IV – Nonsmooth Problems with Constraints

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these slides are under development: please email me for corrections and suggestions



Outline

Introduction

Barrier methods

Penalty and augmented Lagrangian methods

QP with geometric constraints

Introduction

Why Constrained Nonsmooth Optimization?

Many problems in science and engineering are naturally:

- ▶ **Nonconvex**: motion planning, phase retrieval, neural networks
- ▶ **Nonsmooth**: ℓ_1 regularization, matrix rank, contact mechanics
- ▶ **Constrained**: feasibility sets, system dynamics, logical conditions

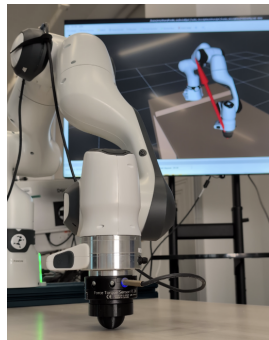
Motivating applications

Signal
recovery

Optimal
control

Machine
learning

Robotics



WILLOW team at INRIA Paris

General Problem Template

Constrained Composite Program

$$\underset{x \in \mathcal{X}}{\text{minimize}} \quad f(x) + g(x) \quad \text{subject to} \quad c(x) \in \mathcal{D}$$

$f : \mathbb{R}^n \rightarrow \mathbb{R}$	smooth
$g : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$	nonsmooth, extended-real-valued
$c : \mathbb{R}^n \rightarrow \mathbb{R}^m$	smooth constraint map
$\mathcal{D} \subset \mathbb{R}^m$	closed set
$\mathcal{X} \subset \mathbb{R}^n$	closed set

Special cases

- ▶ $g = 0$, $\mathcal{X} = \mathbb{R}^n$, $\mathcal{D} = \{0\}$: equality-constrained smooth NLP
- ▶ $g = \delta_C$ (indicator of a convex set C)

Subdifferential Calculus

Convex subdifferential

$$\hat{\partial}\phi(x) := \{s \in \mathbb{R}^n : \phi(y) \geq \phi(x) + \langle s, y - x \rangle, \forall y\}$$

Limiting (Mordukhovich) subdifferential

$$\partial\phi(x) := \{\limsup_{y \rightarrow x} \hat{\partial}\phi(y)\}.$$

Optimality conditions

- ▶ Unconstrained: $0 \in \partial\phi(x^*)$.
- ▶ Constrained ($\min \phi$ s.t. $x \in C$, smooth ϕ): $-\nabla\phi(x^*) \in N_C(x^*)$, with N_C the (limiting) normal cone of C .
- ▶ KKT (smooth NLP): stationarity + complementarity + primal feasibility.

Stationarity notions for nonsmooth nonconvex problems: Clarke, Mordukhovich, proximal, approximately stationary (inexact) ...

Variational analysis, a whole new world...

“Unconstrained” Minimization: Recap

Problem

Given f smooth and g prox-friendly,

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \varphi(x) := f(x) + g(x)$$

Proximal Gradient Step

$$x_{k+1} \in \text{prox}_{\gamma_k g}(x_k - \gamma_k \nabla f(x_k))$$

Derivation: Linearize f around x_k , add proximal term, minimize:

$$x_{k+1} \in \arg \min_x \left\{ f(x_k) + \langle \nabla f(x_k), x - x_k \rangle + g(x) + \frac{1}{2\gamma_k} \|x - x_k\|^2 \right\},$$

check progress, backtrack if needed, repeat.

Approaches to Constraints

Indicator/
exact penalty

Quadratic penalty/
augmented
Lagrangian

Barrier/
interior point

SQP/
composite steps

Each approach makes a different trade-off between

- ▶ anytime feasibility vs. asymptotic feasibility.
- ▶ smoothness/conditioning of subproblems.
- ▶ scalability and per-iteration cost.

Constrained structured optimization

$$\underset{x}{\text{minimize}} \quad f(x) + g(x) \quad \text{subject to} \quad c(x) \leq 0 \quad (\text{P})$$

Assumptions:

- ▶ f and c are smooth
- ▶ g is prox-friendly
- ▶ no convexity assumptions

Applications:

- ▶ (nonsmooth) regularization
- ▶ indicators CC, VC, SC, ...
- ▶ cover $c(x) \in D$ and $g(c(x))$

PIPA *A proximal interior point algorithm with applications to image processing* [CCP20]

IPPROX *An interior proximal gradient method for nonconvex optimization* [DMT24], also [DMT25]

RALMS *An adaptive Lagrangian-based scheme for nonconvex composite optimization* [HT23], also [BST18, CHT22, CT25]

ALPS *Constrained composite optimization and augmented Lagrangian methods* [DMJKM23], also [DM24, DMM24, DMHM25]

KKT-like optimality

Relative to the problem

$$\underset{x}{\text{minimize}} \quad q(x) \quad \text{subject to} \quad c(x) \leq 0$$

a point $\bar{x} \in \mathbb{R}^n$ is called

- ▶ **feasible** if it satisfies $\bar{x} \in \text{dom } q$ and $c(\bar{x}) \leq 0$;
- ▶ **strictly feasible** if it satisfies $\bar{x} \in \text{dom } q$ and $c(\bar{x}) < 0$;
- ▶ **KKT optimal** if there exists $\bar{y} \in \mathbb{R}^m$ such that

$$\begin{aligned} 0 &\in \partial q(\bar{x}) + c'(\bar{x})^\top \bar{y} & c(\bar{x}) &\leq 0 \\ \bar{y} &\geq 0 & \bar{y}_i c_i(\bar{x}) &= 0 \quad \forall i; \end{aligned}$$

- ▶ **ε -KKT optimal** (with $\varepsilon > 0$) if there exists $\bar{y} \in \mathbb{R}^m$ such that

$$\begin{aligned} \text{dist}(-c'(\bar{x})^\top \bar{y}, \partial q(\bar{x})) &\leq \varepsilon & c(\bar{x}) &\leq \varepsilon \\ \bar{y} &\geq 0 & \min\{\bar{y}_i, -c_i(\bar{x})\} &\leq \varepsilon \quad \forall i. \end{aligned}$$

- ▶ **AKKT optimal** if feasible and there exist sequences $\{x^k\}$ and $\{y^k\}$ such that $x^k \rightarrow \bar{x}$, $y^k \geq 0$ and

$$\text{dist}(0, \partial q(x^k) + c'(x^k)^\top y^k) \rightarrow 0, \quad y_i^k c_i(x^k) \rightarrow 0 \quad \forall i;$$

Barrier methods

Structured optimization and barrier idea

We know how to solve

$$\underset{x}{\text{minimize}} \quad \varphi(x) := f(x) + g(x)$$

Now, we may include also a barrier term

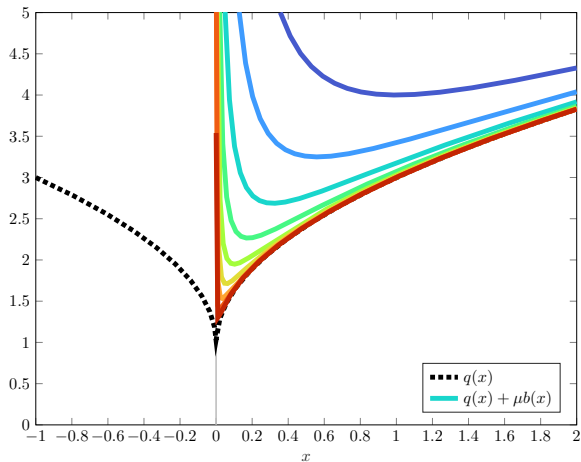
$$\underset{x}{\text{minimize}} \quad \varphi(x) - \log x$$

... to model constraints

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad x \geq 0$$

Illustration

- ▶ smooth term $(f + \mu b)$ has **not full domain!**
- ▶ strong motivation to extend PGM **from globally to locally Lipschitz gradient**



$$x \in \mathbb{R}, f(x) := 1, g(x) := \|x\|_{1/2}^{1/2}, b(x) := 1/x \text{ for } x > 0, \mu \in [0.001, 1]$$

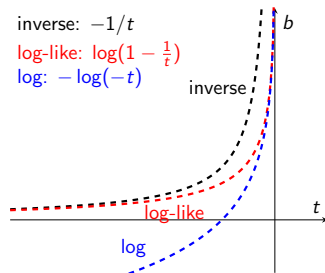
$$\min_x q(x) \text{ s.t. } x \geq 0$$

Interior Point Subproblem

$$\min_x \varphi(x) \quad \text{subject to} \quad c(x) \leq 0 \quad (\text{P})$$

Barrier $b: (-\infty, 0) \rightarrow [0, \infty)$

- ▶ $b' > 0$ on its domain,
- ▶ b convex,
- ▶ $b(t) \rightarrow \infty$ as $t \rightarrow 0^-$.

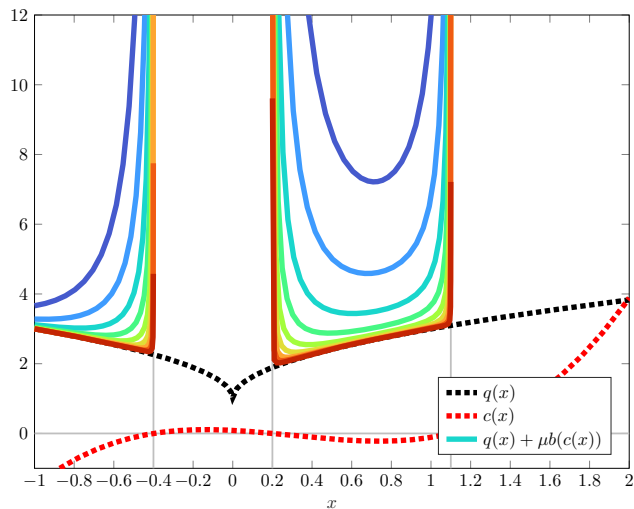


Sequential unconstrained optimization, with barrier parameter $\mu > 0$:

$$\min_x \underbrace{f(x) + \mu \sum_i b(c_i(x))}_{\text{smooth } F_\mu} + g(x) \quad (\text{P}_\mu)$$

The smooth term $F_\mu := f + \mu \sum_i b \circ c_i$ has **not full domain**...

Illustration: barrier subproblems



$x \in \mathbb{R}$, $b(x) := -1/x$ for $x < 0$, $\mu \in [0.001, 1]$, $c(x) = \text{cubic}(x)$

$\min_x q(x)$ s.t. $c(x) \leq 0$

Interior Point Framework

Basic Scheme

$$F_{\mu}(x) := f(x) + \mu \sum_i b(c_i(x))$$

- 1: find a strictly feasible point x^0 for (P)
- 2: choose parameters $\mu_0 > 0$ and $\theta_{\mu} \in (0, 1)$
- 3: **for** $k = 0, 1, 2, \dots$ **do**
- 4: find $x^{k+1} \in \arg \min(F_{\mu_k} + g)$
- 5: set $\mu_{k+1} \leftarrow \theta_{\mu} \mu_k$
- 6: **end for**

$\rightsquigarrow$ ill-conditioning as $\mu \searrow 0$

$\rightsquigarrow$ warm-starting from x_k

$\rightsquigarrow$ affordable subsolver (only descent needed, no global minima)

$\rightsquigarrow$ inexact subsolves

Algorithm: IPPROX [DMT24]

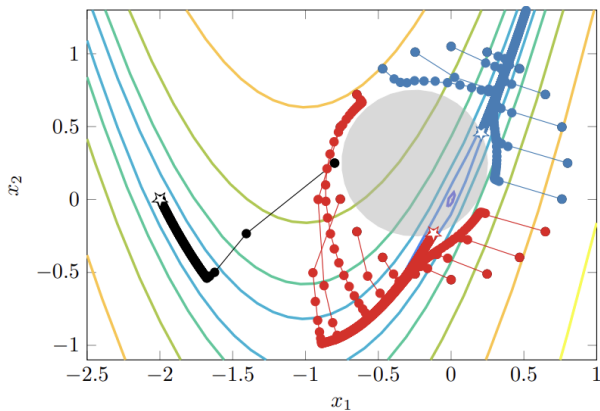
- 1: find a strictly feasible point x^0 for (P)
- 2: select tolerances $\epsilon_p, \epsilon_d > 0$
- 3: choose parameters $\epsilon_0, \mu_0 > 0$ and $\theta_\epsilon, \theta_\mu \in (0, 1)$
- 4: **for** $k = 0, 1, 2, \dots$ **do**
- 5: find an ϵ_k -stationary point x^{k+1} of $F_{\mu_k} + g$ starting from x^k
- 6: set $y_i^{k+1} \leftarrow \mu_k b'(c_i(x^{k+1})) \quad \forall i$
- 7: **if** $\epsilon_k \leq \epsilon_d$ and $\|\min\{y^{k+1}, -c(x^{k+1})\}\| \leq \epsilon_p$ **then**
- 8: **return** $(x^*, y^*) \leftarrow (x^{k+1}, y^{k+1})$
- 9: **end if**
- 10: set $\mu_{k+1} \leftarrow \theta_\mu \mu_k$
- 11: set $\epsilon_{k+1} \leftarrow \max\{\epsilon_d, \theta_\epsilon \epsilon_k\}$
- 12: **end for**

► IPFB: a barrier-friendly proximal gradient method

Nonsmooth Rosenbrock outside a circle [DMT24]

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & 100(x_2 + 1 - (x_1 + 1)^2)^2 + \|x\|_{1/2}^{1/2} \\ \text{s. t.} \quad & \|x - x_c\|^2 \geq r_c^2. \end{aligned}$$

Contour lines of the objective function, circular infeasible set (gray), trajectories of inner and outer iterations for different starting points, and limit points thereof (stars). Trajectories are colored based on the limit point: global (red) and local (blue, black).



Assumptions for IPPROX

$$\min_x \quad \varphi(x) := f(x) + g(x) \quad \text{subject to} \quad c(x) \leq 0 \quad (\text{P})$$

- ▶ $f : \mathbb{R}^n \rightarrow \mathbb{R}$ has locally Lipschitz-continuous gradient.
- ▶ $g : \mathbb{R}^n \rightarrow \overline{\mathbb{R}} := \mathbb{R} \cup \{\infty\}$ is prox-friendly.
- ▶ $c : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has locally Lipschitz-continuous Jacobian.
- ▶ $\inf\{\varphi(x) \mid c(x) \leq 0\} \in \mathbb{R}$.
- ▶ **strict feasibility**, namely

$$\mathcal{F} := \text{dom } \varphi \cap \{x \mid c(x) < 0\} \neq \emptyset.$$

- ▶ some $x^0 \in \mathcal{F}$ is known.

Interior-Point Proximal Method: IPPROX [DMT24]

Finite termination:

For any strictly feasible starting point x^0 and tolerance $\epsilon > 0$, IPPROX returns in finitely many steps an ϵ -KKT optimal point x^* for (P) satisfying $\varphi(x^*) \leq \varphi(x^0)$.

Properties:

- ▶ All iterates are **strictly feasible** ($c(x^k) < 0$) by design
- ▶ As $\mu \rightarrow 0$, minimizers converge to the solution of the original problem
- ▶ Ill-conditioned as $\mu \rightarrow 0$
- ▶ Slow tail convergence...

Key features:

- ▶ **Warm starting**: exploits previous feasible point
- ▶ Objective values **decrease monotonically**
- ▶ **Nonconvex** f : convergence to ϵ -KKT points in finite iterations

We have to deal with equalities too

Can we solve

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad c(x) = 0$$

by means of IPPROX applied to

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad \begin{pmatrix} c(x) \\ -c(x) \end{pmatrix} \leq 0 \quad ?$$

No, because there is no interior / strictly feasible point. We need to relax!

Penalty methods

Indicator and Exact Penalty

Problem:

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad c(x) = 0$$

Indicator instead of constraints? Not really easier...

$$\underset{x}{\text{minimize}} \quad \varphi(x) + \delta_0(c(x))$$

Exact penalty: nonsmooth [Zangwill 1967, Pietrzykowski 1969], often not easy:

$$\underset{x}{\text{minimize}} \quad \varphi(x) + \eta \|c(x)\|_p$$

with $p \in \{1, 2\}$ and $\eta \nearrow \infty$.

Drawback: ill-conditioning for large η ; infeasible iterates.

Forgotten for decades, but now they are coming back [EFOS20, DGO26].

Why? Because ϱ can stay bounded, at least for regular problems— $\bar{\eta} = \|y^*\|_\infty$.

Quadratic Penalty

Problem:

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad c(x) = 0$$

Quadratic penalty yields some smoothness again:

$$\underset{x}{\text{minimize}} \quad \varphi(x) + \varrho \|c(x)\|^2$$

with $\varrho \nearrow \infty$.

Advantage: address subproblem with any reasonable PGM.

$$x_{k+1} \in \arg \min_x \left\{ \underbrace{f(x) + \varrho_k \|c(x)\|^2}_{\text{smooth}} + g(x) \right\}$$

Drawback: ill-conditioning for large ϱ ; infeasible iterates.

Need $\varrho \nearrow \infty$, even for regular problems!

Very often, the quadratic penalty leaves the smooth part with **only locally Lipschitz gradient**. Again, **good motivation for relaxing assumptions in PGM**.

Augmented Lagrangian methods

Augmented Lagrangian Method (ALM)

Problem:

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad c(x) = 0$$

Augmented Lagrangian function

$$\mathcal{L}_\varrho(x, y) := \varphi(x) + \langle y, c(x) \rangle + \frac{\varrho}{2} \|c(x)\|^2$$

ALM scheme

- 1: Given $x^0, y^0, \rho > 0$
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: $x^{k+1} \approx \arg \min_x \mathcal{L}_\varrho(x, y^k)$ *(proximal inner solver)*
- 4: $y^{k+1} = y^k + \varrho c(x^{k+1})$ *(dual ascent)*
- 5: **end for**

Key advantages: warm-start friendly;
multiplier update improves constraint satisfaction at each iteration;
avoids $\varrho \rightarrow \infty$ (for regular problems).

Augmented Lagrangian Method (ALM)

Problem:

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad c(x) = 0$$

Augmented Lagrangian function

$$\mathcal{L}_{\varrho}(x, y) := \varphi(x) + \langle y, c(x) \rangle + \frac{\varrho}{2} \|c(x)\|^2$$

Dual update rule:

$$y^{k+1} = y^k + \varrho c(x^{k+1})$$

Why?

$$\begin{aligned} \partial_x \mathcal{L}_{\varrho_k}(x^{k+1}, y^k) &= \partial \varphi(x^{k+1}) + J_c(x^{k+1})^\top \left(y^k + \varrho_k c(x^{k+1}) \right) \\ &= \partial \varphi(x^{k+1}) + J_c(x^{k+1})^\top y^{k+1} \\ &= \partial_x \mathcal{L}(x^{k+1}, y^{k+1}) \end{aligned}$$

Control **outer** stationarity with **inner** stationarity.

Inexact subsolves with $\varepsilon_k \rightarrow \epsilon_d$ for the minimization of $\mathcal{L}_{\varrho_k}(\cdot, y^k)$.

ALM: Convergence Theory

Convex case [Hes69, Pow69, Roc76]:

- ▶ Global convergence to a KKT point (even with constant ϱ)
- ▶ Convergence rates as for PPA

Nonconvex case [CGT91, BM14, DMJKM23]:

- ▶ Convergence to first-order stationary points (under CQs)
- ▶ Local linear convergence near strict local minima
- ▶ Requires **adaptive** ϱ and **safeguarding** [KS17]

↪ Adaptive ϱ : penalty strong enough.

↪ Safeguarding: multiplier estimate not too crazy.

ALM with adaptive penalty

ALM scheme

```
1: Given  $x^0, y^0, \rho_0 > 0, \kappa > 1, \theta \in (0, 1)$ 
2: for  $k = 0, 1, 2, \dots$  do
3:    $x^{k+1} \approx \arg \min_x \mathcal{L}_{\varrho_k}(x, y^k)$  (proximal inner solver)
4:    $y^{k+1} \leftarrow y^k + \varrho_k c(x^{k+1})$  (dual ascent)
5:   if  $\|c(x^{k+1})\| \leq \max\{\epsilon_p, \theta \|c(x^{k+1})\|\}$  then
6:      $\varrho_{k+1} \leftarrow \varrho_k$ 
7:   else
8:      $\varrho_{k+1} \leftarrow \kappa \varrho_k$  (stronger penalty)
9:   end if
10: end for
```

► $\{\varrho_k\}$ bounded $\implies c(x^k) \rightarrow 0$ (if $\epsilon_p = 0$). Feasible, yey!

► $\varrho_k \nearrow \infty$: since

$$x^{k+1} \in \arg \min_x \varphi(x) + \frac{\varrho_k}{2} \left\| c(x) + \frac{y^k}{\varrho_k} \right\|^2,$$

we **need also** $y^k / \varrho_k \rightarrow 0$ to obtain least-infeasible points. This is **safeguarding**.

ALM with safeguards

ALM iteration — Conn, Gould & Toint [CGT91]

- 1: $x^{k+1} \approx \arg \min_x \mathcal{L}_{\varrho_k}(x, y^k)$ *(proximal inner solver)*
- 2: **if** $\|c(x^{k+1})\| \leq \max\{\epsilon_p, \theta \|c(x^{k+1})\|\}$ **then**
- 3: $y^{k+1} \leftarrow y^k + \varrho_k c(x^{k+1})$, $\varrho_{k+1} \leftarrow \varrho_k$ *(dual ascent)*
- 4: **else**
- 5: $\varrho_{k+1} \leftarrow \kappa \varrho_k$, $y^{k+1} \leftarrow y^k$ *(stronger penalty)*
- 6: **end if**

ALM iteration — Birgin & Martinez [BM14]

- 1: $\hat{y}^k \leftarrow \text{proj}_Y(y^k)$, given a compact set Y
- 2: $x^{k+1} \approx \arg \min_x \mathcal{L}_{\varrho_k}(x, \hat{y}^k)$ *(proximal inner solver)*
- 3: $y^{k+1} \leftarrow y^k + \varrho_k c(x^{k+1})$ *(dual ascent)*
- 4: **if** $\|c(x^{k+1})\| \leq \max\{\epsilon_p, \theta \|c(x^{k+1})\|\}$ **then**
- 5: $\varrho_{k+1} \leftarrow \varrho_k$
- 6: **else**
- 7: $\varrho_{k+1} \leftarrow \kappa \varrho_k$ *(stronger penalty)*
- 8: **end if**

Constrained Structured Programs [DMJKM23]

Problem (P)

$$\underset{x}{\text{minimize}} \quad \varphi(x) \quad \text{subject to} \quad c(x) \in \mathcal{D}$$

$\varphi := f + g$, f and c smooth, g nonsmooth, $\mathcal{D}$ closed simple set.

How can we use all the **ALM machinery for equality constraints**?

Just add some variables and split!

$$\begin{aligned} &\underset{x,z}{\text{minimize}} && \varphi(x) + \delta_{\mathcal{D}}(z) \\ &\text{subject to} && c(x) - z = 0 \end{aligned}$$

Low-rank matrix completion [DMT25]

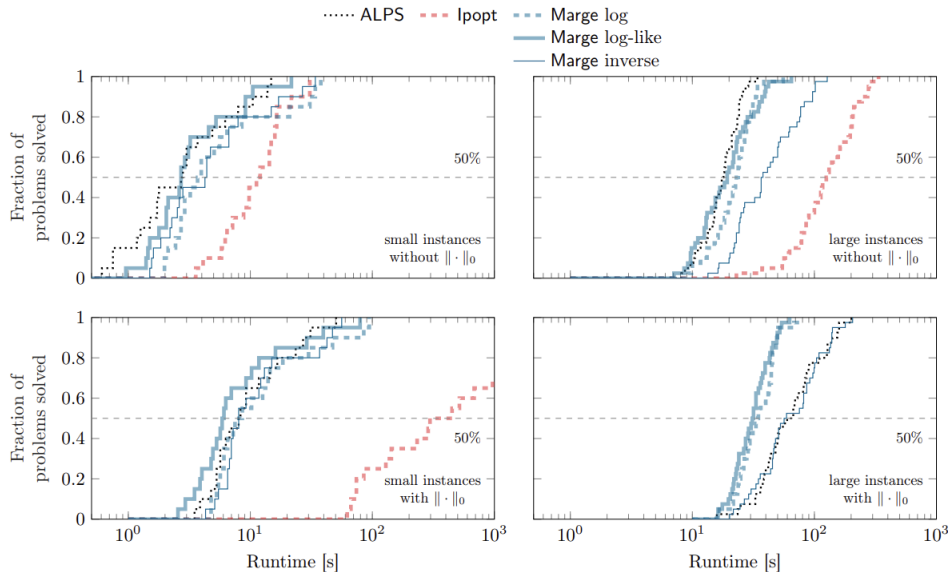
We have an incomplete matrix Y . We want a low-rank, dense approximation $X \approx Y$ within a range $[Y_{\min}, Y_{\max}]$. Use $X \equiv UV^\top$.

$$\begin{aligned} \underset{U, V}{\text{minimize}} \quad & \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} (Y_{i,j} - U_{i,:} V_{j,:}^\top) + \frac{\lambda}{n_m} \sum_{j=1}^{n_m} \|V_{j,:}\|_0 \\ \text{w. r. t.} \quad & U \in \mathbb{R}^{n_u \times n_a}, V \in \mathbb{R}^{n_m \times n_a} \\ \text{s. t.} \quad & \max\{Y_{\min}, Y_{i,j} - 1\} \leq U_{i,:} V_{j,:}^\top \leq \max\{Y_{\max}, Y_{i,j} + 1\} \quad \forall (i,j) \in \Omega \\ & Y_{\min} \leq U_{i,:} V_{j,:}^\top \leq Y_{\max} \quad \forall (i,j) \notin \Omega \\ & \|U_{i,:}\| = 1 \quad \forall i = 1, \dots, n_u \end{aligned}$$

Problem size: $n = n_a(n_u + n_m)$ variables and $m = 2n_u n_m$ inequality constraints.

MovieLens dataset (movie ratings) $\rightsquigarrow n \in [1790, 9930]$, $m \in [2130, 38920]$.

Low-rank matrix completion [DMT25]



Ipopt (red thick dashed lines) is not included on large instances with $\|\cdot\|_0$ because of excessive runtime.

QP with geometric constraints: exploiting structure

Geometric / Structured Constraints

Many practical constraints have special **structure**:

- ▶ **Polyhedral**: boxes, simplices, linear equalities/inequalities
- ▶ **Conic**: second-order cone, PSD matrices
- ▶ **Nonconvex**: cardinality, rank, logical/combinatorial conditions
- ▶ **Geometric (set membership)**: $x \in \mathcal{C}$, $\mathcal{C}$ possibly nonconvex

Problem

$$\underset{x}{\text{minimize}} \quad \frac{1}{2}x^\top Qx + q^\top x \quad \text{subject to} \quad Ax \in \mathcal{C}$$

$Q \succeq 0$, $\mathcal{C}$ closed (possibly nonconvex), projection onto $\mathcal{C}$ available.

Examples for $\mathcal{C}$:

- ▶ Box + cardinality: $\{x : \|x\|_0 \leq s, l \leq x \leq u\}$ (sparse solutions)
- ▶ Union of sets (logical conditions, integer-like)
- ▶ Polytopes (control limits, allocation constraints)

Why this problem?

Convex quadratic programs (QPs) with polyhedral constraints are standard in signal processing, control, and decision-making.

Real applications need more: **Disjunctive constraints**

- ▶ *Logical conditions* — e.g. at most one of two actuators fires at a time
- ▶ *Cardinality / sparsity constraints* — e.g. portfolio selection
- ▶ *Obstacle avoidance* — feasible region is a union of halfspaces

These constraints are **nonconvex**, but projections onto them are tractable.

Constraint type	$\mathcal{C}$
Box / polyhedron	$\mathcal{C} = \{z : \ell \leq z \leq u\}$
Complementarity	$\mathcal{C} = \{z : z_1 \geq 0, z_2 \geq 0, z_1 z_2 = 0\}$
Either-or (EOC)	$\mathcal{C} = \{z : z_1 \leq u_{\max}, z_2 \leq u_{\max}, z_1 z_2 = 0\}$
Cardinality	$\mathcal{C} = \{z : \ z\ _0 \leq s\}$

The Problem Class

Geometric QP

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2}x^\top Qx + q^\top x \quad \text{subject to} \quad Ax \in \mathcal{C} \quad (\text{P})$$

- ▶ $Q \succeq 0$, $A \in \mathbb{R}^{m \times n}$, closed (possibly nonconvex) set $\mathcal{C} \subset \mathbb{R}^m$.
- ▶ Only requirement on $\mathcal{C}$: a **projection oracle** $\text{proj}_{\mathcal{C}}$ is available.

Question: Can we exploit quadratic cost and projections onto $\mathcal{C}$ efficiently?

Strategy: augmented Lagrangian and structure-exploitation with

- ▶ subproblem reformulation via condensing [ZAB⁺19, DM25]
- ▶ splitting and subproblem solver [MBJ⁺25]

Reduce computational cost significantly while preserving ALM convergence guarantees.

Augmented Lagrangian (AL) Approach

Introduce slack $z \in \mathcal{C}$, split (P) as:

$$\underset{x, z}{\text{minimize}} \quad f(x) + \delta_{\mathcal{C}}(z) \quad \text{subject to} \quad Ax - z = 0$$

The (proximal) **augmented Lagrangian** function

$$\mathcal{L}_{\varrho}(x, z; \hat{x}, \hat{y}) := f(x) + \hat{y}^{\top} (Ax - z) + \frac{\varrho}{2} \|Ax - z\|^2 + \frac{\sigma}{2} \|x - \hat{x}\|^2$$

Algorithm: Safeguarded ALM

Repeat until $\|Ax_k - z_k\|_{\infty} \leq \epsilon_p$ and stationarity $\varepsilon_k \leq \epsilon_d$:

1. **Solve** $(x_k, z_k) \approx \arg \min_{x, z} \mathcal{L}_{\varrho_k}(x, z, \hat{x}_k, \hat{y}_k)$ s.t. $z \in \mathcal{C}$
2. **Update** dual $y_{k+1} \leftarrow \hat{y}_k + \varrho_k(Ax_k - z_k)$
3. Adapt parameters $\sigma_{k+1}, \varrho_{k+1}$ and tolerance ε_{k+1}

Guarantees: feasible accumulation points are KKT points; infeasible ones are stationary for the feasibility measure.

Subproblem Structure and Condensing

$$\mathcal{L}_\varrho(x, z; \hat{x}, \hat{y}) := f(x) + \hat{y}^\top (Ax - z) + \frac{\varrho}{2} \|Ax - z\|^2 + \frac{\sigma}{2} \|x - \hat{x}\|^2$$

For fixed $z \in \mathcal{C}$, the subproblem is *unconstrained* and *strictly convex* in x :

$$\begin{aligned} X_k(z) &:= \arg \min_x \mathcal{L}_{\varrho_k}(x, z; \hat{x}_k, \hat{y}_k) \\ &= \arg \min_x \left\{ f(x) + \hat{y}_k^\top (Ax - z) + \frac{\varrho_k}{2} \|Ax - z\|^2 + \frac{\sigma_k}{2} \|x - \hat{x}_k\|^2 \right\} \end{aligned}$$

$$0 = Qx + q + A^\top \hat{y}_k + \varrho_k A^\top (Ax - z) + \sigma_k (x - \hat{x}_k)$$

Solved by a **single linear system** (matrix independent of z , factorize once):

$$\underbrace{(Q + \sigma_k \mathbb{I} + \varrho_k A^\top A)}_{\succ 0, \text{ const. w.r.t. } z} x = \sigma_k \hat{x}_k - q + A^\top (\varrho_k z - \hat{y}_k)$$

The **condensed subproblem** is then $\min_{z \in \mathcal{C}} V_k(z)$ with the **marginal function** $V_k := z \mapsto \mathcal{L}_{\varrho_k}(X_k(z), z; \hat{x}_k, \hat{y}_k)$.

V_k is smooth and easy to compute!

What Condensing Achieves

Dimension reduction

Variables: from $\mathbb{R}^{n+m}$ to $\mathbb{R}^m$

Built-in preconditioning

Construction preconditions the subproblem automatically

Efficient gradient

$\nabla V_k(z)$ is a cheap backsolve away

Solver-agnostic

Any proximal method applies to $\text{minimize}_{z \in \mathcal{C}} V_k(z)$

Hard equality constraints

Lower-level $A_{\text{eq}}x = b_{\text{eq}}$ kept *exactly* in condensed step via KKT system:

$$\begin{pmatrix} Q + \sigma_k \mathbb{I} & A^\top & A_{\text{eq}}^\top \\ A & -\varrho_k \mathbb{I} & 0 \\ A_{\text{eq}} & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ \lambda \\ \lambda_{\text{eq}} \end{pmatrix} = \begin{pmatrix} \dots \\ \dots \\ b_{\text{eq}} \end{pmatrix}$$

Retains sparsity and symmetry; avoids ill-conditioning from penalizing well-structured constraints

Numerical results

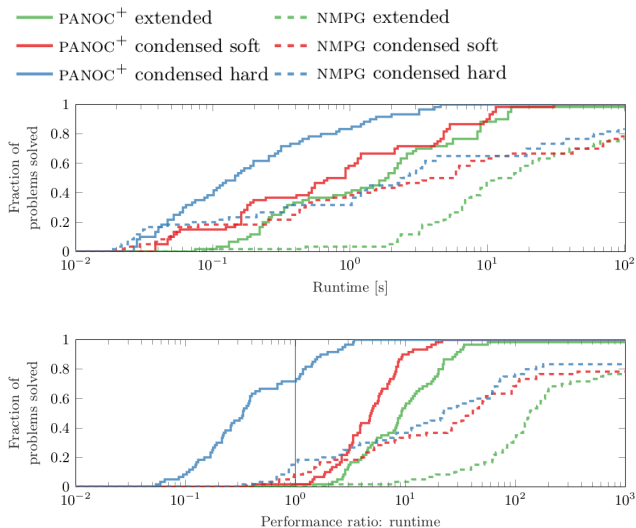
Obstacle benchmark: Discretized 1-D Laplace with obstacle set:

$$\begin{aligned} & \underset{x_x, x_y, x_z \in \mathbb{R}^N}{\text{minimize}} && \frac{1}{2} \|x_x\|^2 + \frac{1}{2} \|x_y\|^2 - \mathbf{1}^\top x_y \\ & \text{subject to} && x_x \geq 0, \quad 0 \leq x_y \perp x_z \geq 0 \\ & && x_x + Ax_y = x_z. \end{aligned}$$

Degenerate solution $x_x = x_y = x_z = 0$.

Six solver configurations: NMPG and PANOC⁺, extended and condensed, soft and hard. $\varepsilon_p = \varepsilon_d = 10^{-6}$; 100 s time limit

- ▶ Scales to $N = 256$; [JKMW23] caps at $N = 64$.
- ▶ Iters reduced from *millions* to *thousands* using panoc+.
- ▶ Degeneracy handled robustly.

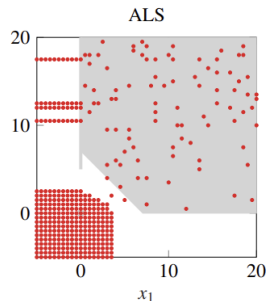
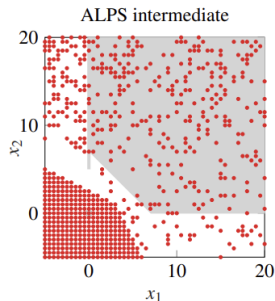
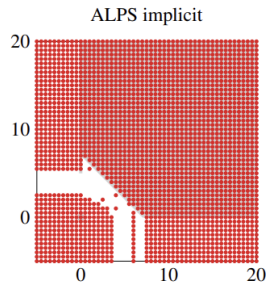
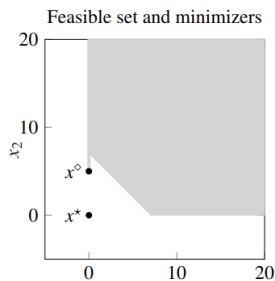


▶ condensed-hard $\succ$ condensed-soft $\succ$ extended.

Problem with Vanishing Constraints [Hoh09, DM24]

$$\begin{aligned} & \underset{x \in \mathbb{R}^2}{\text{minimize}} && 4x_1 + 2x_2 \\ & \text{s. t.} && x_1 \geq 0, \quad x_2 \geq 0 \\ & && x_1 > 0 \implies x_1 + x_2 \geq 5\sqrt{2} \\ & && x_2 > 0 \implies x_1 + x_2 \geq 5 \end{aligned}$$

Comparison of solutions obtained using different methods and formulations. Feasible set (gray background) and grid of (2601 uniformly spaced) starting points and corresponding termination mark depending on the returned solution: global minimizer $x^* = (0, 0)$ (red circle) or local minimizer $x^\diamond = (0, 5)$ (omitted mark).



More variables, more splitting: divide and conquer

Geometric QP

$$\underset{x}{\text{minimize}} \quad \frac{1}{2}x^\top Qx + q^\top x \quad \text{subject to} \quad x \in \mathcal{X}, \quad Ax \in \mathcal{C} \quad (\text{P})$$

Let's add $z \in \mathcal{C}$

$$\underset{x,z}{\text{minimize}} \quad f(x) + \delta_{\mathcal{X}}(x) + \delta_{\mathcal{C}}(z) \quad \text{subject to} \quad Ax = z$$

and then $(\tilde{x}, \tilde{z}) \in \mathcal{E} := \{(x, z) \mid Ax = z\}$

$$\underset{x,z,\tilde{x},\tilde{z}}{\text{minimize}} \quad f(\tilde{x}) + \delta_{\mathcal{E}}(\tilde{x}, \tilde{z}) + \delta_{\mathcal{X}}(x) + \delta_{\mathcal{C}}(z) \quad \text{subject to} \quad x = \tilde{x}, \quad z = \tilde{z}$$

Menager et al.'s "Frictional Contact-Implicit Inverse Dynamics" (2025)

What now? ALM

$$\underset{x, z, \tilde{x}, \tilde{z}}{\text{minimize}} \quad f(\tilde{x}) + \delta_{\mathcal{E}}(\tilde{x}, \tilde{z}) + \delta_{\mathcal{X}}(x) + \delta_{\mathcal{C}}(z) \quad \text{subject to} \quad x = \tilde{x}, \quad z = \tilde{z}$$

Augmented Lagrangian function

$$\begin{aligned} \mathcal{L}_{\rho}(x, z, \tilde{x}, \tilde{z}, y_x, y_z) &:= f(\tilde{x}) + \delta_{\mathcal{E}}(\tilde{x}, \tilde{z}) + \delta_{\mathcal{X}}(x) + \delta_{\mathcal{C}}(z) \\ &\quad + y_x^{\top}(x - \tilde{x}) + y_z^{\top}(z - \tilde{z}) + \frac{\rho}{2}\|x - \tilde{x}\|^2 + \frac{\rho}{2}\|z - \tilde{z}\|^2 \end{aligned}$$

AL subproblem

$$\underset{x, z, \tilde{x}, \tilde{z}}{\text{minimize}} \quad \mathcal{L}_{\rho}(x, z, \tilde{x}, \tilde{z}, \hat{y}_x, \hat{y}_z)$$

How do we minimize $\mathcal{L}_{\rho}(\cdot, \cdot, \cdot, \cdot, \hat{y}_x, \hat{y}_z)$?

We can call proximal-gradient methods

but, luckily, we can apply **alternating minimization**!

Alternating subsolver: (x, z)

AL subproblem

$$\underset{x, z, \tilde{x}, \tilde{z}}{\text{minimize}} \quad \mathcal{L}_\varrho(x, z, \tilde{x}, \tilde{z}, \hat{y}_x, \hat{y}_z)$$

with respect to (x, z) becomes

$$\begin{aligned} \arg \min_{x, z} \quad & y_x^\top (x - \tilde{x}) + y_z^\top (z - \tilde{z}) + \frac{\varrho}{2} \|x - \tilde{x}\|^2 + \frac{\varrho}{2} \|z - \tilde{z}\|^2 \quad \text{subject to} \quad x \in \mathcal{X}, \quad z \in \mathcal{C} \\ \iff \arg \min_{x, z} \quad & \frac{\varrho}{2} \|x - \tilde{x} + y_x / \varrho\|^2 + \frac{\varrho}{2} \|z - \tilde{z} + y_z / \varrho\|^2 \quad \text{subject to} \quad x \in \mathcal{X}, \quad z \in \mathcal{C} \\ \iff \quad & x \in \text{proj}_{\mathcal{X}}(\tilde{x} - y_x / \varrho), \quad z \in \text{proj}_{\mathcal{C}}(\tilde{z} - y_z / \varrho) \end{aligned}$$

Alternating subsolver: $(\tilde{x}, \tilde{z})$

AL subproblem

$$\underset{x, z, \tilde{x}, \tilde{z}}{\text{minimize}} \quad \mathcal{L}_\varrho(x, z, \tilde{x}, \tilde{z}, \hat{y}_x, \hat{y}_z)$$

with respect to $(\tilde{x}, \tilde{z})$ becomes

$$\arg \min_{\tilde{x}, \tilde{z}} \quad f(\tilde{x}) + y_x^\top (x - \tilde{x}) + y_z^\top (z - \tilde{z}) + \frac{\varrho}{2} \|x - \tilde{x}\|^2 + \frac{\varrho}{2} \|z - \tilde{z}\|^2 \quad \text{subject to} \quad A\tilde{x} = \tilde{z}$$

$$\iff \begin{cases} 0 = Q\tilde{x} + q - y_x + \varrho(\tilde{x} - x) + A^\top \lambda \\ 0 = -y_z + \varrho(\tilde{z} - z) - \lambda \\ A\tilde{x} = \tilde{z} \end{cases}$$

$$\iff \begin{bmatrix} Q + \varrho \mathbb{I} & \cdot & A^\top \\ \cdot & \varrho \mathbb{I} & -\mathbb{I} \\ A & -\mathbb{I} & \cdot \end{bmatrix} \begin{pmatrix} \tilde{x} \\ \tilde{z} \\ \lambda \end{pmatrix} = \begin{pmatrix} -q + y_x + \varrho x \\ y_z + \varrho z \\ 0 \end{pmatrix}$$

$$\iff \begin{bmatrix} Q + \varrho \mathbb{I} & A^\top \\ A & -\frac{1}{\varrho} \mathbb{I} \end{bmatrix} \begin{pmatrix} \tilde{x} \\ \lambda \end{pmatrix} = \begin{pmatrix} -q + y_x + \varrho x \\ z + \frac{1}{\varrho} y_z \end{pmatrix}$$

Alternating minimization subsolver

Instead of invoking PGM (with associated stepsizes and backtrack), we can

$$\underset{x, z, \tilde{x}, \tilde{z}}{\text{minimize}} \quad \mathcal{L}_\rho(x, z, \tilde{x}, \tilde{z}, \hat{y}_x, \hat{y}_z)$$

by selecting an initial pair $(\tilde{x}, \tilde{z})$ and then repeating the steps

► $(x, z) \leftarrow \arg \min \mathcal{L}_\rho(\cdot, \cdot, \tilde{x}, \tilde{z}, \hat{y}_x, \hat{y}_z),$

► $(\tilde{x}, \tilde{z}) \leftarrow \arg \min \mathcal{L}_\rho(x, z, \cdot, \cdot, \hat{y}_x, \hat{y}_z),$

over and over again until convergence to a stationary point of $\mathcal{L}_\rho(\cdot, \cdot, \cdot, \cdot, \hat{y}_x, \hat{y}_z)$.

Simple steps, no need to call an inner prox-grad solver!

QPCC for robotics

$$\begin{aligned} & \underset{x_a, x_c, x_\sigma}{\text{minimize}} && \frac{1}{2} \|Ax_a + Bx_c + b\|^2 \\ & \text{subject to} && x_\sigma = Cx_a + Dx_c + g \\ & && \mathcal{K}_\mu \ni x_c \perp x_\sigma + \Gamma(x_\sigma) \in \mathcal{K}_\mu^* \end{aligned}$$

tracking cost

kinematics / dynamics

frictional contact

$$(0 \leq x_c \perp x_\sigma \geq 0)$$

Quadrupod

our method

Ipopt

homotopy Ipopt

Tower

our method

Ipopt

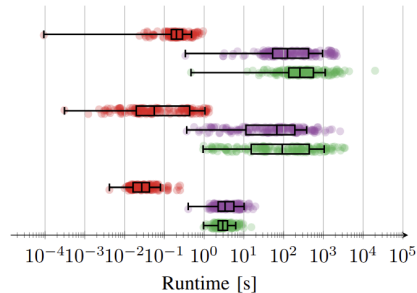
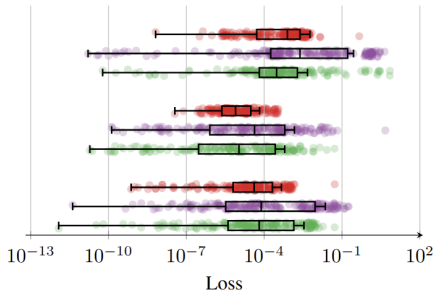
homotopy Ipopt

Trunk

our method

Ipopt

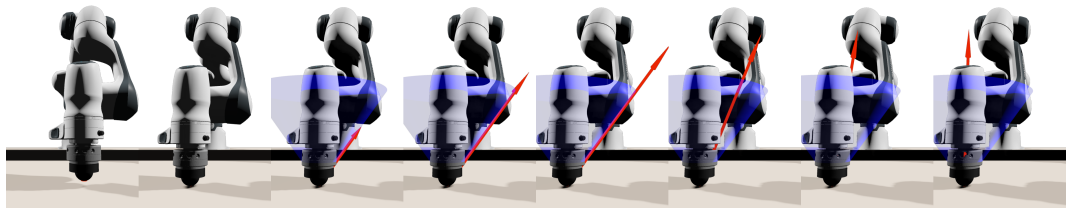
homotopy Ipopt



Comparison on NCP problems across robots, considering a nonlinear programming method (Ipopt), a regularized scheme with homotopy, and the alternating-ALM approach. [MBJ⁺25]

Frictional contact-implicit inverse dynamics

$$\begin{aligned} & \underset{x_a, x_c, x_\sigma}{\text{minimize}} && \frac{1}{2} \|Ax_a + Bx_c + b\|^2 \\ & \text{subject to} && x_\sigma = Cx_a + Dx_c + g \\ & && \mathcal{K}_\mu \ni x_c \perp x_\sigma + \Gamma(x_\sigma) \in \mathcal{K}_\mu^* \end{aligned}$$



WILLOW team at INRIA Paris

Franka sanding task: contact cone (blue) and ground reaction force (red arrow). The robot first makes contact, then slide, and comes to a rest. [MBJ⁺25]

[go2 handstand]

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